NEHAL QADEER

AI & Automation Engineer | Python · GenAI · Computer Vision

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Work authorization: Post-study residence permit

SUMMARY

AI and automation engineer with an M.Sc. in Applied Computer Science (SRH Heidelberg, 2025). I build GenAI applications, AI agents, computer vision and workflow automations in Python, including a multi-agent RAG system (94.5% faithfulness) and real-time object detection at 25+ FPS on standard CPUs.

TECHNICAL SKILLS

AI & GenAI: LLM integration, prompt engineering, RAG (hybrid search, reranking), multi-agent systems, LLM evaluation (RAGAS), PII redaction (Presidio), Google Gemini API

Computer Vision: YOLOv8, Intel OpenVINO, OpenCV, PyTorch, PyQt6

Languages & APIs: Python, SQL, JavaScript, TypeScript · FastAPI, REST APIs, JSON, webhooks, OAuth 2.0, SQLAlchemy

Automation & Data: Make.com, Zapier, Apify, Selenium · PostgreSQL, MySQL, MongoDB, Redis

Tools & Methods: Git/GitHub, Docker, Docker Compose, Pytest, Linux/Bash, coding agents (Claude Code, GitHub Copilot) · Agile/Scrum, requirements and user stories, technical documentation

PROFESSIONAL EXPERIENCE

Teleoperation & Data Labeling Intern | Sereact GmbH (Robotics AI), Stuttgart *Oct 2026 – Present*

- Record robot imitation-learning demonstrations via teleoperation and quality-check each episode before it is used for AI model training.
- Debug Python and Bash data-collection scripts, isolate edge cases and write structured incident reports for engineering.

Project Manager | ZELLE Solutions, Karachi (hybrid) *Oct 2022 – Nov 2023*

- Led a cross-functional team of 7 developers and UI/UX designers through Agile/Scrum delivery, from requirements to deployment.
- Ran stand-ups and stakeholder meetings, refined requirements into user stories, wrote REST API documentation and coordinated release QA.

Brand Development Executive | ZELLE Solutions, Karachi (hybrid) *May 2022 – Oct 2022*

- Analyzed user behavior data and campaign performance metrics, and ran competitor and market analyses.

Customer Support Representative | Breakthru (ApexChat), Karachi *Jun 2020 – Oct 2022*

- Supported 3,000+ customers, handling 350 to 450 chats per 7.5-hour shift across 3 to 4 parallel conversations within KPI targets.
- Documented every case in the eCRM system to support quality reviews and reporting.

EDUCATION

M.Sc. Applied Computer Science | SRH Hochschule Heidelberg · Grade 2.2 *2023 – 2025*

- Focus: Software Engineering, Distributed Systems, Cloud Architectures.

B.Sc. Computer Science | IQRA University, Karachi *2015 – 2019*

- Algorithms, data structures, object-oriented programming (Python, basics of C#), relational database systems.

PROJECTS

Multi-Agent RAG System for Regulated Environments (GDPR, EU AI Act) | *Python*, *FastAPI*, *Docker Compose*, *Presidio*, *RAGAS*

- Designed a 4-agent pipeline (planner, hybrid retriever, verifier, synthesizer) that answers questions from internal documents with source citations.
- Added cross-encoder reranking, PII removal before indexing and a tamper-proof audit log.
- Measured answer quality with an automated RAGAS evaluation suite: 94.5% faithfulness, plus relevance and recall.

Object Detection Game for Hearing-Impaired Children (Master's Thesis) | *YOLOv8*, *Intel OpenVINO*, *OpenCV*, *PyQt6*

- Trained a custom YOLOv8 model and optimized it with OpenVINO to run at 25+ FPS on standard laptop CPUs, without a GPU.
- Created an accessible desktop game that shows detected objects as large visual labels for children who cannot rely on audio.

Virtual Room Builder & German Vocabulary Game | *FastAPI*, *React*, *Google Gemini Vision*, *PostgreSQL*, *Pytest*

- Developed a full-stack learning app with a FastAPI backend, PostgreSQL with Alembic migrations and a 60 FPS canvas room editor.
- Integrated Gemini Vision to turn photos into structured vocabulary cards, with spoken pronunciation and a 15/15 passing Pytest suite.

Workflow Automation & Data Pipelines | *Python*, *Selenium*, *Apify*, *Make.com*, *Zapier*, *PostgreSQL*

- Deployed scrapers as scheduled cloud jobs collecting 2,000 to 4,000 records per run, validated in Make.com and loaded into PostgreSQL.
- Cut a client's 40-hour manual lead-research task to 1 hour, and built a 4-app Zapier automation (Forms, Sheets, Calendar, Gmail) with OAuth.

LANGUAGES & ADDITIONAL

Languages: English (full professional proficiency, IELTS) · German (A1, currently learning)
Additional: Operations Associate, Amazon, Germany (Jul 2026 – Sep 2026) · Delivery Rider, Mannheim (Jan 2026 – Jul 2026)